Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/16/2022
PI Name:	Hayman née Anderson, Allison Ph.D.		
Project Title:	Multi-Environment Virtual Training for Long Duration Exploration Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-5004	**Congressional District:**	2
Comments:	NOTE: name change to Hayman née Anderson (Ed., March 2025). PI moved to University of Colorado from Dartmouth College in early 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	05/19/2021	**End Date:**	05/18/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	We have added an additional team member, Dr. Daniel Szafir, as Co-I from the University of North Carolina. He is an Assistant Professor in the Department of Computer Science. His research emphasizes the use of VR for human-robot interaction. In particular, his team will collaborate on the development of the surface EVA task.		
COI Name (Institution):	Clark, Torin Ph.D. (University of Colorado, Boulder) Robinson, Stephen Ph.D. (University of California, Davis) Vance, Eric Ph.D. (University of Colorado, Boulder) Szafir, Daniel (University of North Carolina)		
Grant/Contract No.:	80NSSC21K1140		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of adaptive VR for training in long duration exploration missions (LDEM). These missions are likely to have long time periods between pre-mission training and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry decent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (EVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth prior to the start of the mission, and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission critical LDEM tasks. We will experimentally investigate how skills transfer from the training environment to a physical setting, the degree to which the training environment enables skills maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to un-trained tasks. As a secondary benefit, we will investigate training in an adaptive, complex VR environment as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological response in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. The novel contributions of this research are to investigate the use of immersive, adaptive VR environments to train astronauts specifically designed for LDEMs by 1) Incorporating multiple, mission critical tasks for a mission to Mars; 2) Investigating skills transfer, retention, and generalizability; 3) Designing with low volume, mass, and operational overhead; and 4) Investigating the degree to which the proposed training can also provide secondary benefits of broad brain stimulation. Our proposed research is directly aligned with that of the Virtual NASA Specialized Center of Research (VNSCOR), and can be integrated with researchers investigating teams, optimal adaptive control algorithms, motivational strategies, or neural and cognitive decrements with both human and animal studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has wide application in training in immersive environments, particularly for complex tasks. We develop a novel training algorithm to facilitate rapid skills acquisition, while also ensuring broad improvement in performance (rather than unidimensional performance). To our knowledge is absent from the literature, but may improve how we administer training in an automated (rather than supervised) training paradigm. This research will also contribute to our understanding of immersive environments and the degree to which they stimulate the brain. This informs our scientific understanding of neuroplasticity when learning and acquiring new skills. This has applications in rehabilitation, to inform how using these environments may yield better patient outcomes.
Task Progress:	Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of VR for training in long duration exploration missions (LDEM). These missions are likely to have long time periods between pre-mission training, and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry descent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (EVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth pre-mission and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission critical LDEM tasks. We will investigate how skills transfer from the VR training environment to a physical setting, the degree to which the training environment enables performance maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to untrained tasks. As a secondary benefit, training in an adaptive, complex environment could serve as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological response in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. In Year 1, we have begun developing an LDEM-specific, multi-environment virtual training simulator, called Trinity. Three environments focused on Mars mission concepts are incorporated into Trinity: EDL, habitat maintenance/repair, and EVA using a surface rover. It will be presented using a head-mounted-display (HMD, HTC Vive Pro). Task difficulty in Trinity will adaptively scale with user-specific acquisition of skills. We have developed a novel algorithm philosophy to enable automated training for complex tasks. To facilitate training, we have developed our server infrastructure to track subject progress on tasks. The training algorithm and data storage have been implemented in our EDL task, which we have also dramatically improved in fidelity and capability over Year 1. We are currently in the process of pilot testing this environment. We will assess skills transfer going from the virtual training environment to a high-fidelity test environment (physical mock-up) in human subject experiments. We have obtained our Institutional Review Board (IRB) approval for this experiment. Our physical space cockpit simulator has been upgraded to include an out-the-window view of the Martian terrain, and we have worked with a University of Colorado Boulder (CU Boulder) graduate projects team to develop a human-rated Martian rover cockpit. We have also begun investigating the utility of Trinity as a countermeasure to neural decrements in spaceflight. We will quantify brain activation during training using functional near-infrared spectroscopy (fNIRS) and electroencephalogram (EEG) -- for which we have begun developing our protocols for both devices. We anticipate that an immersive, spaceflight-relevant trainer will stimulate more regions of the brain than non-immersive cognitive tasks without mission relevance. Finally, we have put forth considerable effort for Virtual NASA Specialized Center of Research (VNSCOR) integration with the Neo and Morpheus teams, resulting in the presentation and submission of our Exit Definition Phase documents. In future years, Aims include investigating the use of the virtual training environment for performance maintenance, and demonstrating long-term acceptability, engagement, and skills transfer in an isolated, confined, controlled environment with other researchers in the Mars Adaptive TRaining-Integrative Knowledge System (MATRIKS) VNSCOR.
Bibliography Type:	Description: (Last Updated: 02/18/2026)
Abstracts for Journals and Proceedings	Anderson A, Boppana A, Putman E, Lewis Q, Peterson B, Clark T. "Trinity virtual reality training environment and adaptive training algorithm for long duration human spaceflight." 92nd Aerospace Medical Association Meeting, Reno, NV, May 22-26, 2022. Abstracts. 92nd Aerospace Medical Association Meeting, Reno, NV, May 22-26, 2022. , May-2022

Abstracts for Journals and Proceedings	Boppana A, Putman E, Anderson A. "Dynamic virtual reality training algorithm for long duration exploration missions. " 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022 (Poster). , Feb-2022
Abstracts for Journals and Proceedings	Putman E, Boppana A, Clark T, Anderson A. "Multi-environment adaptive virtual reality training as a potential countermeasure for spaceflight associated neuroplasticity." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022 (Poster). , Feb-2022
Abstracts for Journals and Proceedings	Anderson A, Boppana A, Putman E, Lewis Q, Peterson B, Clark T. "Trinity: Multi-environment virtual trainer for long duration exploration missions." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022 (Poster). , Feb-2022